

Physics 340 Final Exam: Mathematical Physics Solution Set

Question 1: Algebra of Vectors

Given

$$\mathbf{A} = \langle 2, -1, 4 \rangle, \quad \mathbf{B} = \langle -1, 3, 2 \rangle.$$

a) **Dot Product**

$$\mathbf{A} \cdot \mathbf{B} = 2(-1) + (-1)(3) + 4(2) = -2 - 3 + 8 = 3.$$

$$\boxed{\mathbf{A} \cdot \mathbf{B} = 3}$$

b) **Cross Product**

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 4 \\ -1 & 3 & 2 \end{vmatrix}$$

$$= \hat{i}((-1)(2) - 4(3)) - \hat{j}(2(2) - 4(-1)) + \hat{k}(2(3) - (-1)(-1))$$

$$= -14\hat{i} - 8\hat{j} + 5\hat{k}.$$

$$\boxed{\mathbf{A} \times \mathbf{B} = \langle -14, -8, 5 \rangle}$$

c) **Angle Between the Vectors**

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos \theta.$$

$$|\mathbf{A}| = \sqrt{21}, \quad |\mathbf{B}| = \sqrt{14}.$$

$$\cos \theta = \frac{3}{\sqrt{21}\sqrt{14}} = \frac{3}{\sqrt{294}}.$$

$$\theta = \cos^{-1}\left(\frac{3}{\sqrt{294}}\right) \approx 79.9^\circ$$

Question 2: Coupled Linear First-Order Differential Equations

$$\frac{dx}{dt} = 4x + y, \quad \frac{dy}{dt} = 2x + 3y.$$

a) **Matrix Form**

Let

$$\vec{X} = \begin{pmatrix} x \\ y \end{pmatrix}.$$

Then

$$\frac{d\vec{X}}{dt} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} \vec{X}.$$

$$\boxed{A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}}$$

b) **Eigenvalues and Eigenvectors**

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{vmatrix} = (4 - \lambda)(3 - \lambda) - 2$$

$$= 12 - 7\lambda + \lambda^2 - 2 = \lambda^2 - 7\lambda + 10$$

$$(\lambda - 5)(\lambda - 2) = 0$$

$$\boxed{\lambda_1 = 5, \quad \lambda_2 = 2}$$

For $\lambda_1 = 5$:

$$A - 5I = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}$$

$$-x + y = 0 \Rightarrow y = x$$

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda_2 = 2$:

$$A - 2I = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$$

$$2x + y = 0 \Rightarrow y = -2x$$

$$\vec{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

c) **Solve for** $x(t)$, $y(t)$

General solution:

$$\vec{X}(t) = C_1 e^{5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^{2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

So

$$x(t) = C_1 e^{5t} + C_2 e^{2t}, \quad y(t) = C_1 e^{5t} - 2C_2 e^{2t}$$

Use $x(0) = 1$, $y(0) = 2$:

$$C_1 + C_2 = 1$$

$$C_1 - 2C_2 = 2$$

Subtracting gives

$$-3C_2 = 1 \Rightarrow C_2 = -\frac{1}{3}$$

Then

$$C_1 = \frac{4}{3}$$

Therefore

$$x(t) = \frac{4}{3}e^{5t} - \frac{1}{3}e^{2t}$$

$$y(t) = \frac{4}{3}e^{5t} + \frac{2}{3}e^{2t}$$

d) **Behavior of the System**

Both eigenvalues are positive, so solutions grow exponentially away from the origin. Since the eigenvalues are real and distinct, trajectories do not spiral.

The dominant long-term behavior is governed by e^{5t} , so trajectories align with the eigenvector $\langle 1, 1 \rangle$.

The origin is an unstable node.

Question 3: AC RC Circuit and Complex Impedance

$$V(t) = V_0 e^{i\omega t}$$

A resistor R and capacitor C are connected in series.

a) **Total Impedance**

$$Z_R = R, \quad Z_C = \frac{1}{i\omega C} = -\frac{i}{\omega C}$$

Thus

$$Z = R - \frac{i}{\omega C}$$

b) **Current**

Using $V = IZ$,

$$I(t) = \frac{V_0 e^{i\omega t}}{R - \frac{i}{\omega C}}$$

Magnitude of impedance:

$$|Z| = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

So current amplitude is

$$I_0 = \frac{V_0}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$$

c) **Phase Shift**

$$\phi_Z = -\tan^{-1}\left(\frac{1}{\omega CR}\right)$$

Therefore current has phase

$$\phi = \tan^{-1}\left(\frac{1}{\omega CR}\right)$$

Current leads voltage.

d) **Physical Meaning**

The resistor dissipates energy and produces no phase shift. The capacitor stores energy in the electric field, causing current to respond before voltage reaches maximum.

Question 4: Fourier Analysis

$$f(x) = \begin{cases} -1, & -L < x < 0, \\ 1, & 0 < x < L, \end{cases} \quad \text{period } 2L$$

Use

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

a) **Coefficients**

Since $f(x)$ is odd:

$$\boxed{a_0 = 0, \quad a_n = 0}$$

Now compute b_n :

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Because odd $\times$ odd = even,

$$b_n = \frac{2}{L} \int_0^L \sin\left(\frac{n\pi x}{L}\right) dx$$

Let

$$k = \frac{n\pi}{L}$$

Then

$$\begin{aligned} b_n &= \frac{2}{L} \left[-\frac{\cos(kx)}{k} \right]_0^L \\ &= \frac{2}{L} \cdot \frac{1 - \cos(n\pi)}{k} \\ &= \frac{2}{L} \cdot \frac{1 - (-1)^n}{n\pi/L} \end{aligned}$$

$$\boxed{b_n = \frac{2(1 - (-1)^n)}{n\pi}}$$

So

$$b_n = \begin{cases} \frac{4}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

b) **Fourier Series**

$$f(x) = \frac{4}{\pi} \left[\sin\left(\frac{\pi x}{L}\right) + \frac{1}{3} \sin\left(\frac{3\pi x}{L}\right) + \frac{1}{5} \sin\left(\frac{5\pi x}{L}\right) + \dots \right]$$

c) **Convergence**

At points where $f(x)$ is continuous, the Fourier series converges to $f(x)$.

At discontinuities, it converges to the midpoint of the jump:

$$\frac{f(x^-) + f(x^+)}{2}$$

At $x = 0$:

$$\frac{-1 + 1}{2} = 0$$

Likewise at $x = \pm L$, the series also converges to 0.

Continuous points: $f(x)$, jump points: 0

Question 5: Vector Calculus

$$\mathbf{F} = (xy)\hat{i} + (x^2 + z)\hat{j} + (y + z^2)\hat{k}$$

a) **Divergence**

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(x^2 + z) + \frac{\partial}{\partial z}(y + z^2)$$

$$= y + 0 + 2z$$

$$\boxed{\nabla \cdot \mathbf{F} = y + 2z}$$

b) **Curl**

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ xy & x^2 + z & y + z^2 \end{vmatrix}$$

$$= 0\hat{i} + 0\hat{j} + x\hat{k}$$

$$\boxed{\nabla \times \mathbf{F} = x\hat{k}}$$

c) **Line Integral**

$$\mathbf{r}(t) = \langle t, t^2, t \rangle, \quad 0 \leq t \leq 1$$

$$\mathbf{r}'(t) = \langle 1, 2t, 1 \rangle$$

Substitute:

$$x = t, \quad y = t^2, \quad z = t$$

$$\mathbf{F}(\mathbf{r}(t)) = \langle t^3, t^2 + t, 2t^2 \rangle$$

Dot product:

$$\mathbf{F} \cdot \mathbf{r}' = t^3 + (t^2 + t)(2t) + 2t^2$$

$$= 3t^3 + 4t^2$$

Therefore

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (3t^3 + 4t^2) dt$$

$$= \frac{3}{4} + \frac{4}{3} = \frac{25}{12}$$

$$\boxed{\int_C \mathbf{F} \cdot d\mathbf{r} = \frac{25}{12}}$$